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Predictive Modeling of Cardiovascular Disease Using Logistic Regression and Statistical Analysis

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Abstract

Cardiovascular disease (CVD) is a leading global health concern,contributing to millions of deaths annually. This study aims to identify key risk factors associated with CVD and to develop a predictive model using statistical tools. Data comprising 70,000 records with 11 explanatory variables were obtained from Kaggle. Statistical techniques including logistic regression, chi-square tests, and Mann-Whitney U tests were used to analyze the relationship between CVD and potential predictors such as age, blood pressure, cholesterol, glucose level, smoking status, alcohol consumption, and physical activity. The logistic regression model achieved an accuracyof 72.14%, identifying age, systolic blood pressure, cholesterol, and physical activity as significantpredictors.The findings underscore the importance of early detection and lifestyle management in reducing CVD risk.

Keywords: Cardiovascular Disease, Logistic Regression, Risk Factors, Predictive Modeling